CSE471

Deep Learning for Computer Vision

Lecture 2: Image Fundamentals and Image Preprocessing

Lovely Professional University

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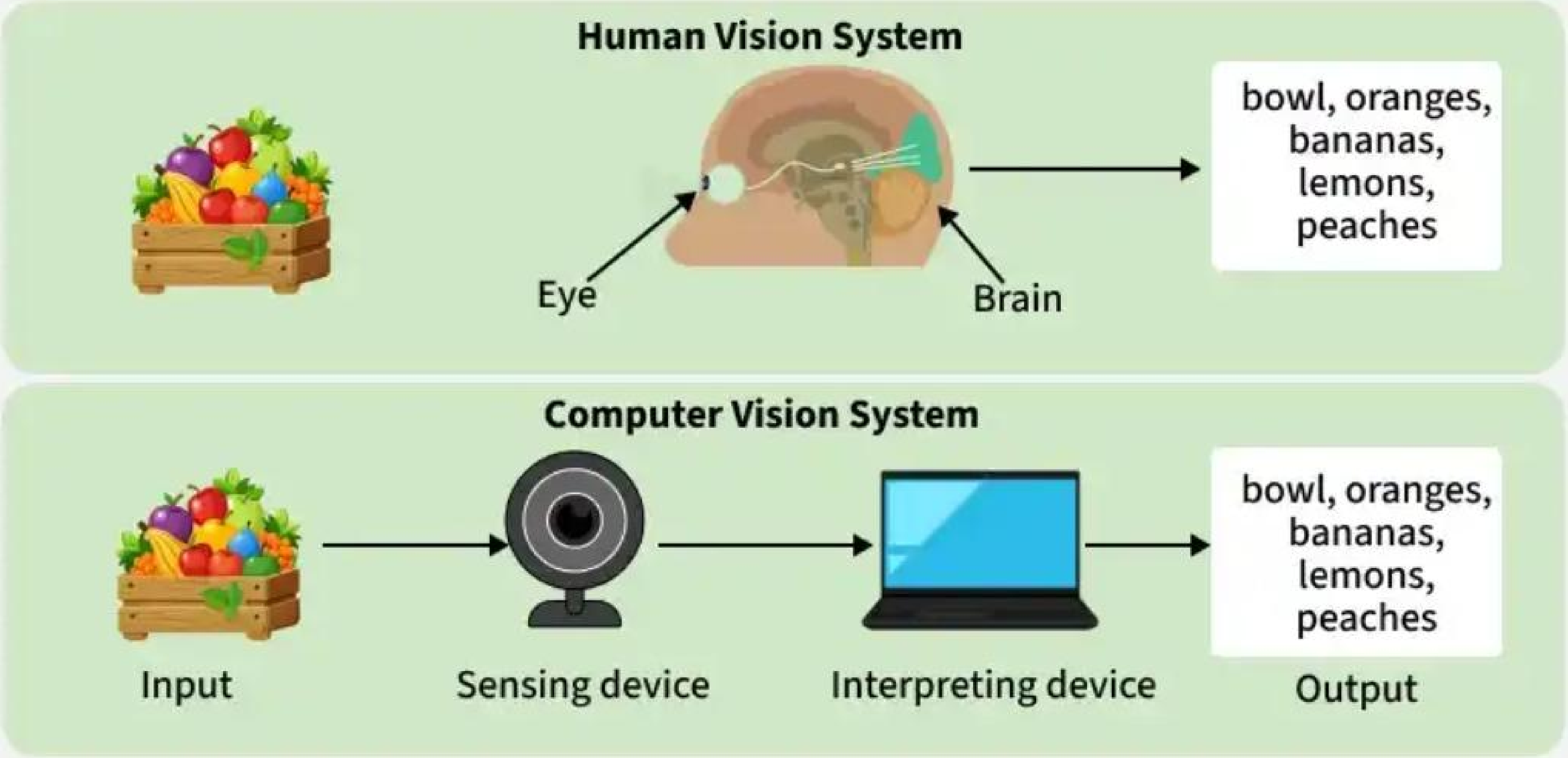
Learning Outcomes

By the end of this lecture, you should be able to:

- Explain the composition of **digital images** as numerical matrices.
- Understand pixels, intensity values, and image resolution.
- Differentiate between grayscale and RGB formats.
- Explain various **color spaces** and their applications.
- Perform foundational **image preprocessing** operations.



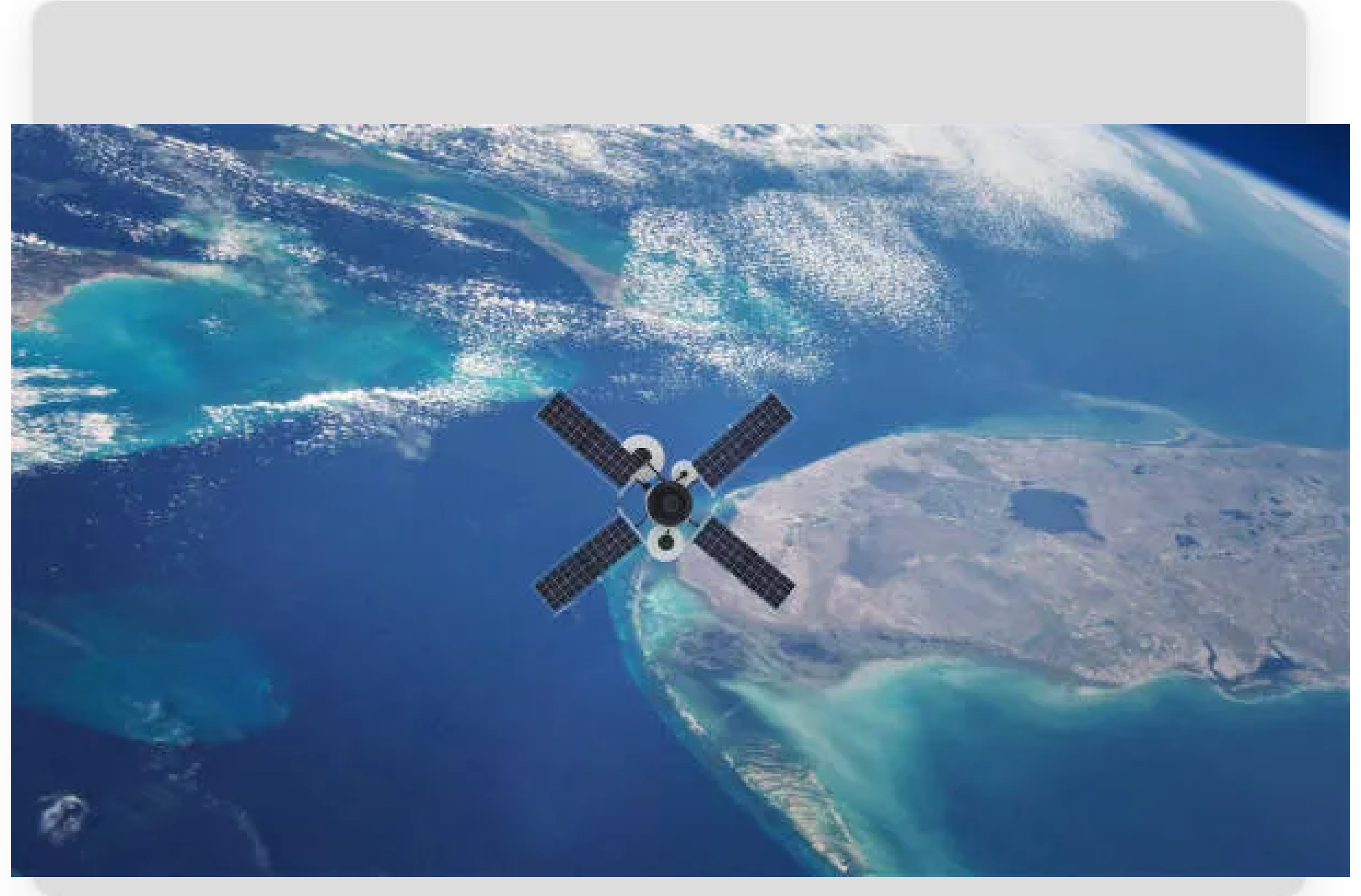
Lecture Roadmap

Digital Images → Pixels & Resolution → Channels & Color Spaces → Image Histograms

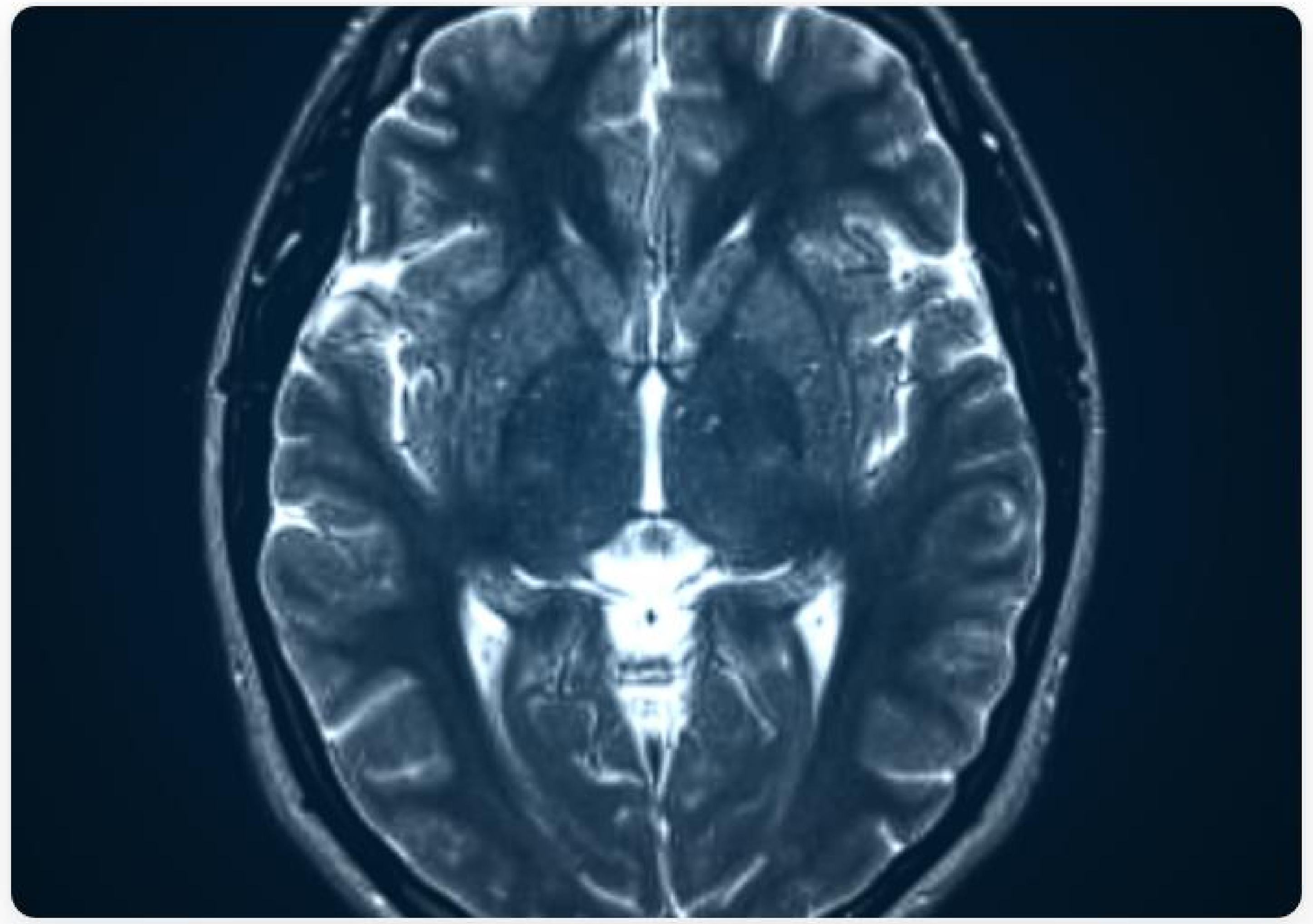
Image Preprocessing → Data Augmentation → Pipeline Example → Summary

What do all these have in common?

Selfie



Satellite Image



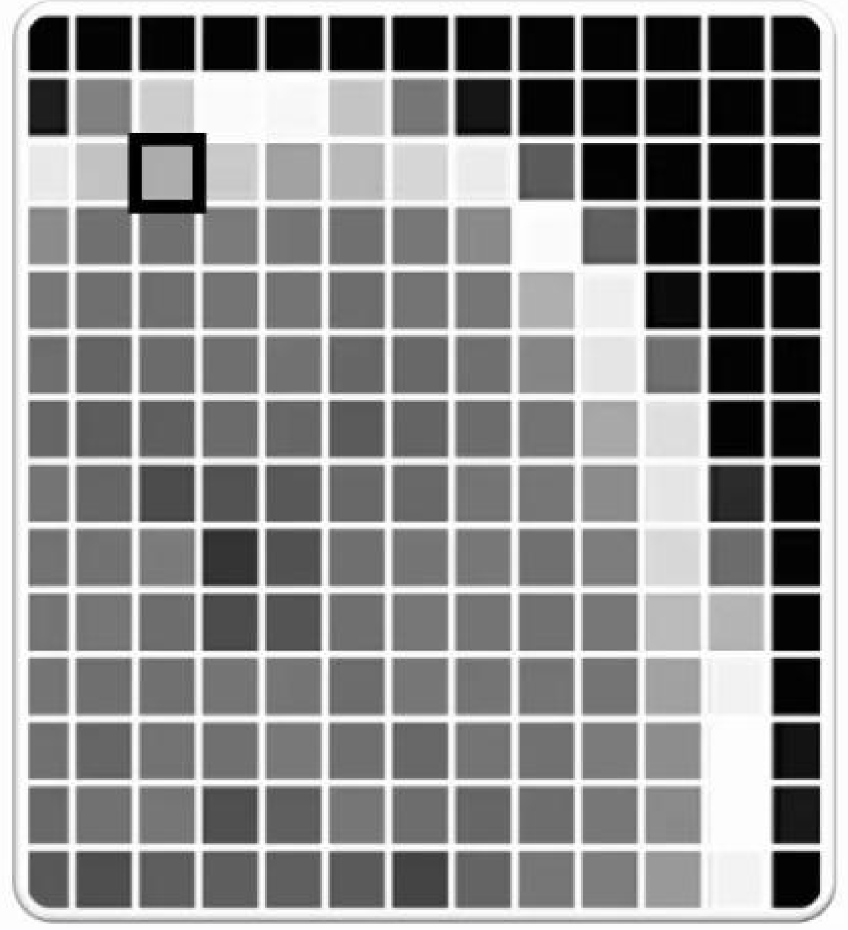
MRI Scan

They are all just matrices of numbers.

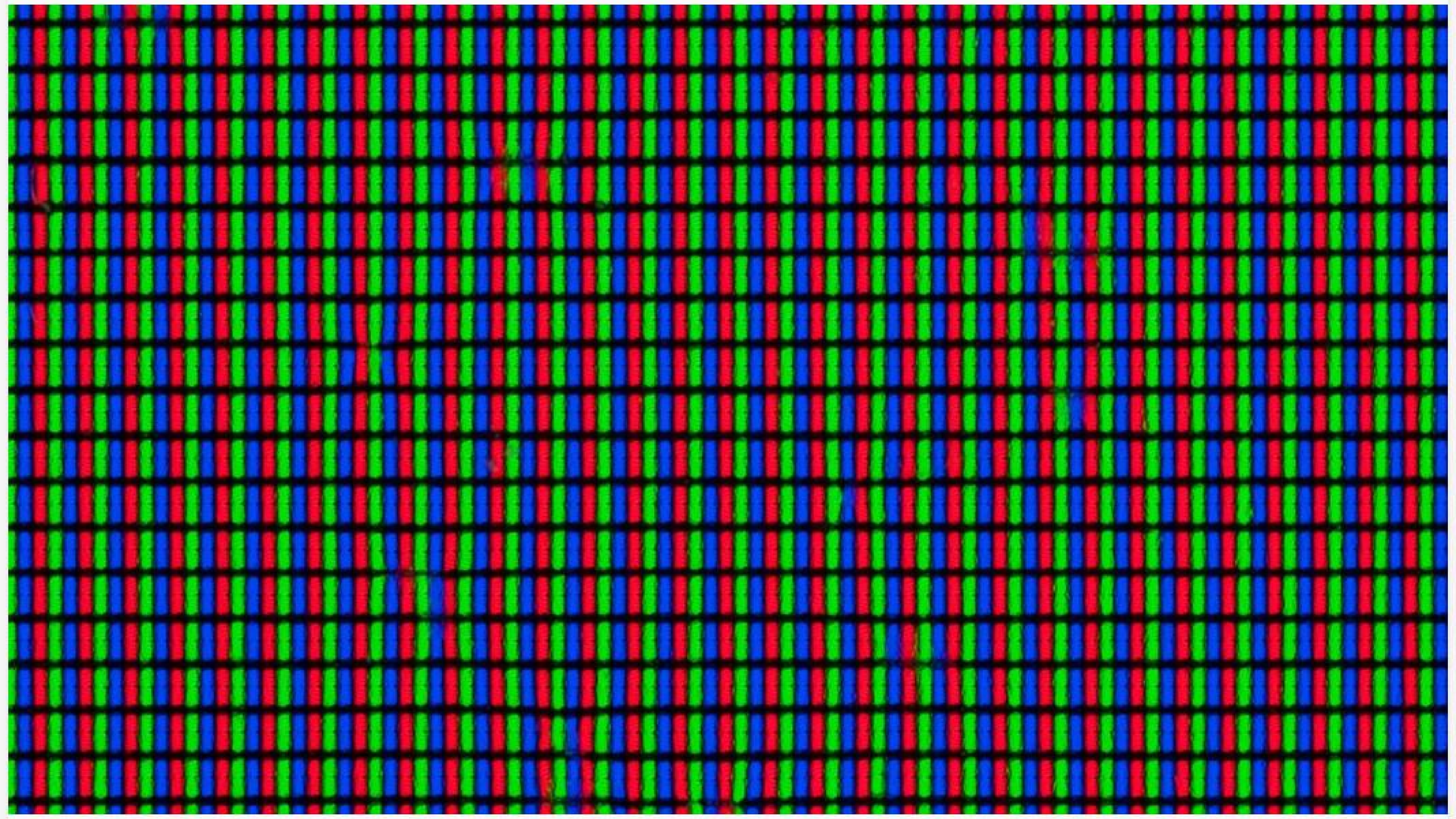
What is a Digital Image?

A digital image is a numeric representation of a two-dimensional image.

- **Continuous to Discrete:** Real-world light is continuous; cameras sample it into a discrete grid.
- **The Matrix:** Formally, an image I is a 2D function $f(x, y)$.
- The spatial coordinates (x, y) map to a specific Pixel.
- The amplitude of f at any pair of coordinates is called the **intensity**.



Understanding Pixels

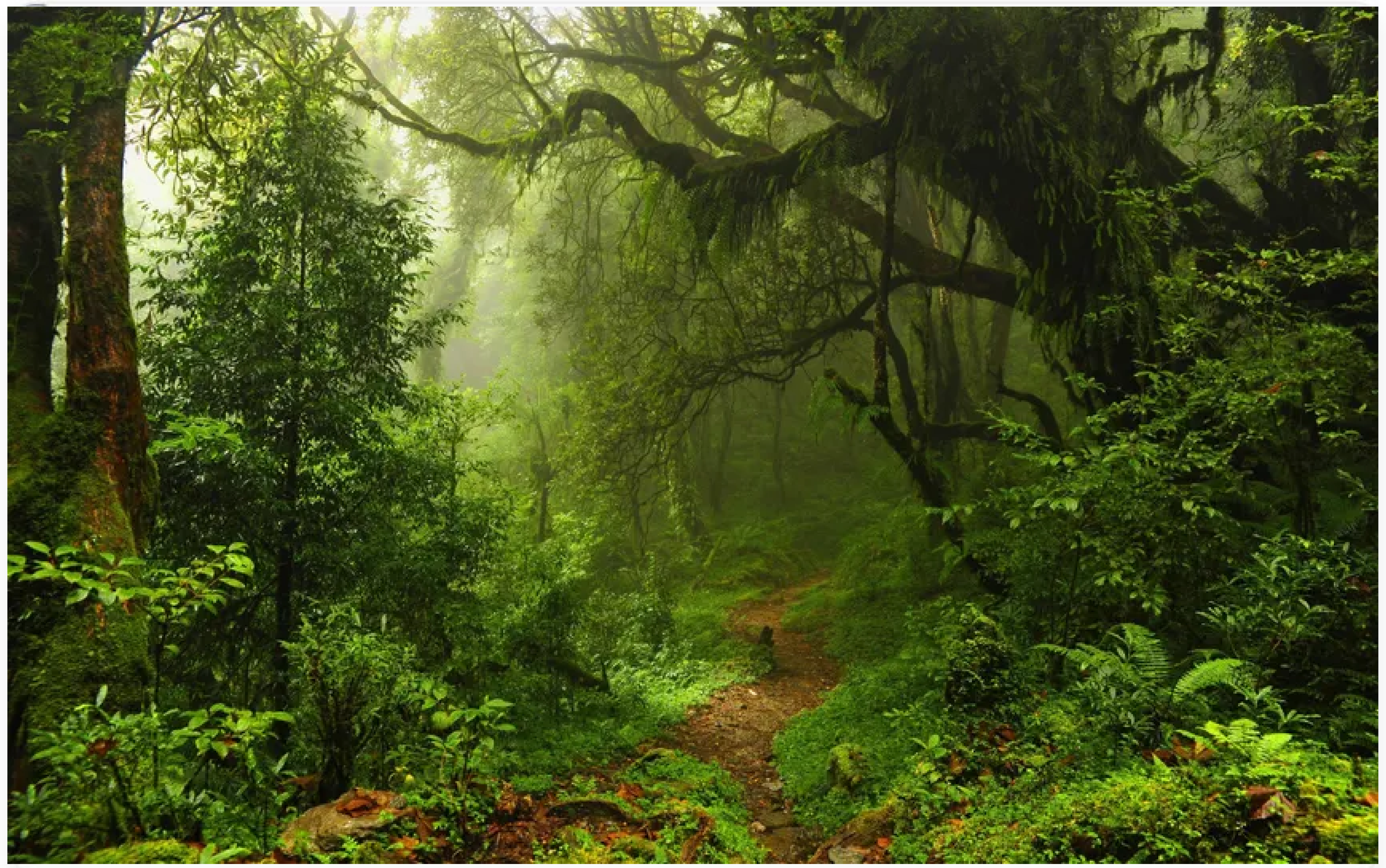


Pixel stands for *Picture Element*. It is the smallest addressable element in a raster image.

- **Coordinates:** Usually defined as (x, y) with $(0, 0)$ at the top-left corner.
- **Intensity / Brightness:** In standard 8-bit images, intensity ranges from 0 to 255.
- **Color Value:** Hardware (like the monitor on the left) uses tiny Red, Green, and Blue subpixels to create millions of colors.

Image Resolution

Low Resolution (e.g., 640×480)



High Resolution (e.g., 3840×2160)

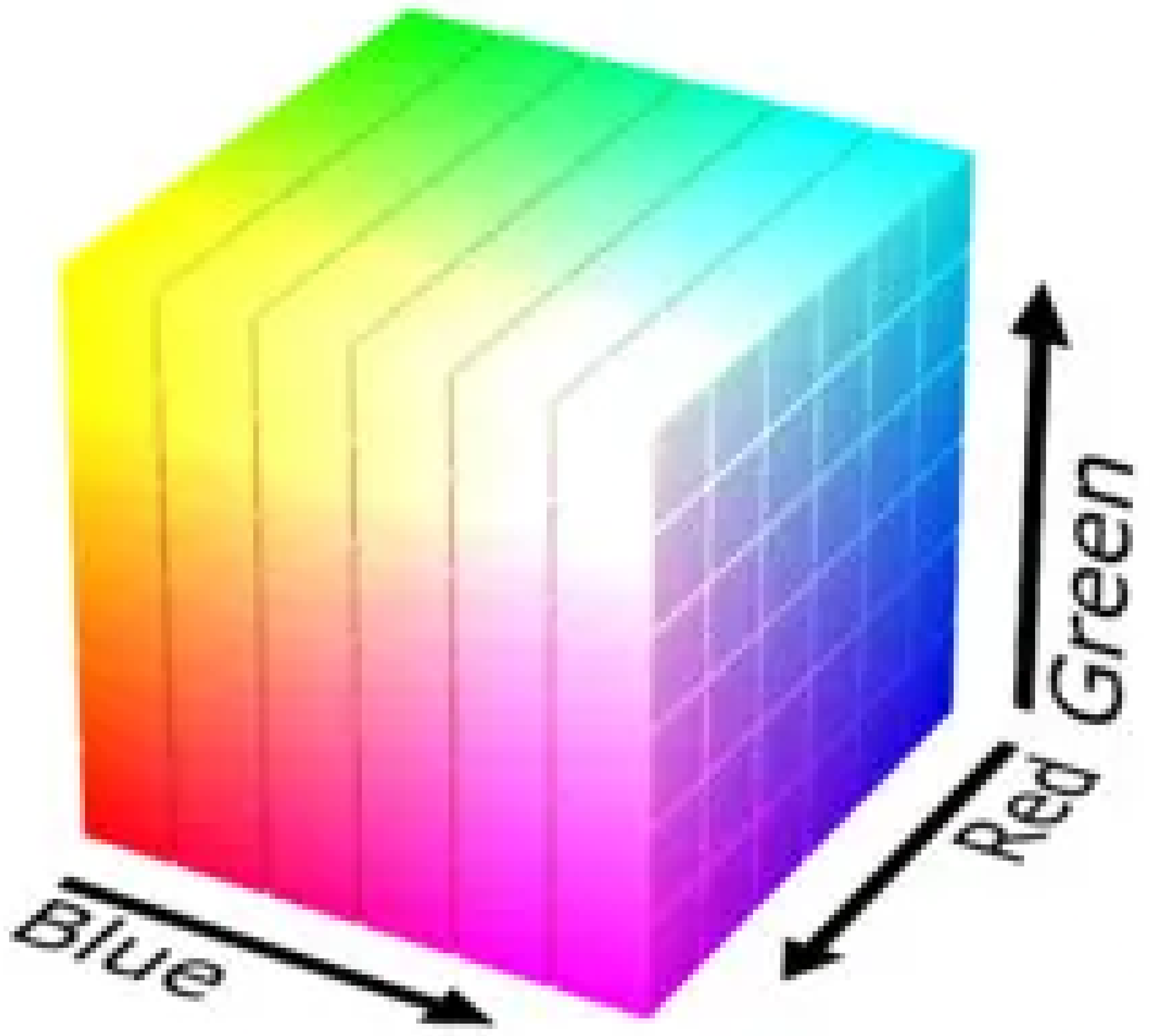
Higher Res → More Detail → More Memory → Longer Processing Time

Image Channels

A channel is a 2D array of pixels. Images can be composed of multiple stacked channels.

- **Grayscale:** 1 Channel (Intensity only). Shape: $(H, W, 1)$
- **RGB:** 3 Channels (Red, Green, Blue). Shape: $(H, W, 3)$
- **RGBA:** 4 Channels (RGB + Alpha/Transparency).
- **Multispectral:** Used in satellite/medical imaging (e.g., 10+ channels mapping infrared, UV).

Color Spaces



Different mathematical models to represent color tuples.

- **RGB (Red, Green, Blue):** Default for displays. Cube geometry.
- **BGR:** Used by OpenCV by default (legacy camera format).
- **HSV (Hue, Saturation, Value):** Cylinder geometry. Excellent for color detection (separates color from brightness).
- **LAB / YCbCr:** Separates luminance from chrominance. Used for image enhancement and compression (JPEG).

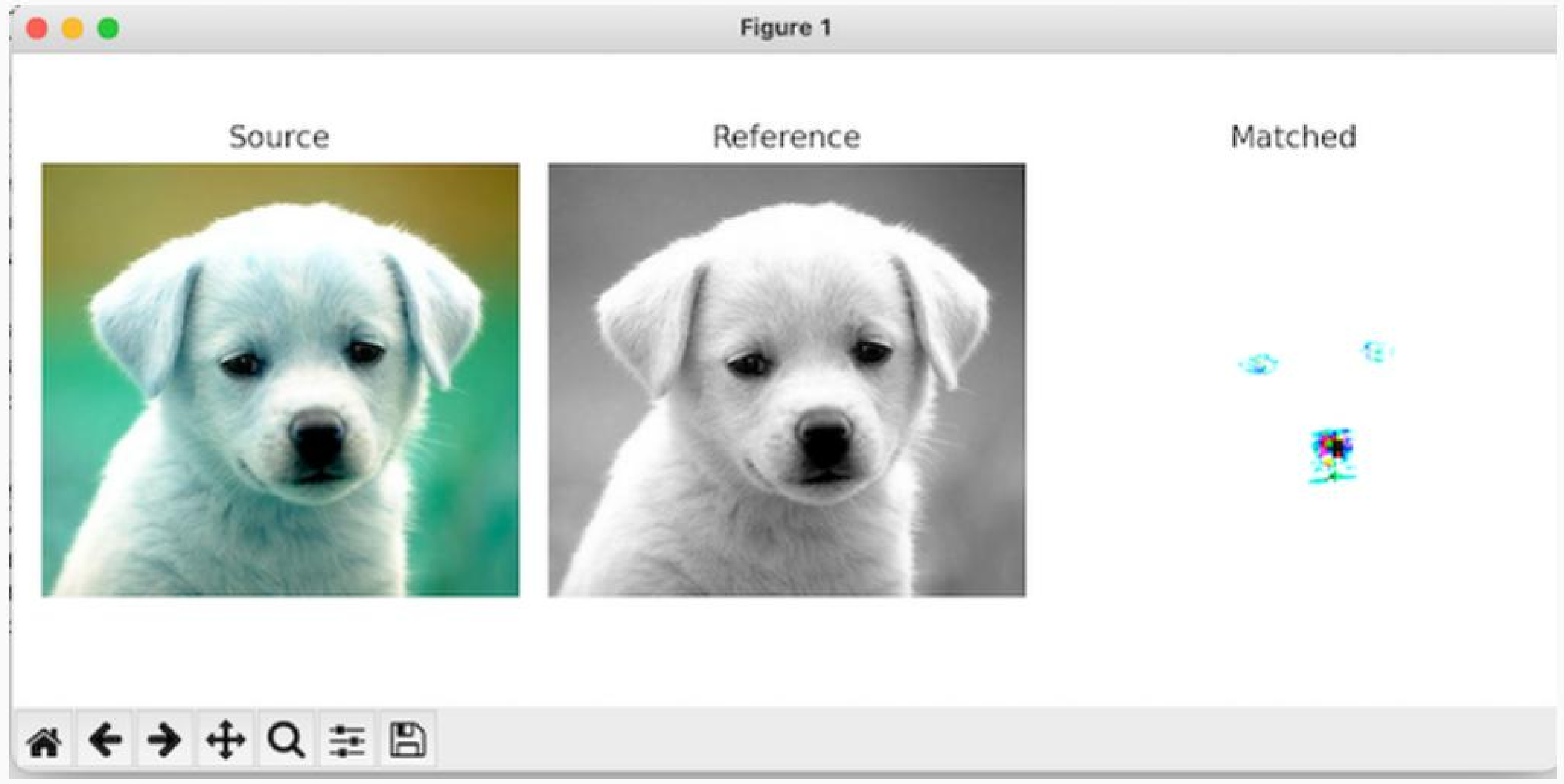
Image File Formats Comparison

Format	Compression	Transparency	Typical Application in CV
JPEG / JPG	Lossy (high efficiency)	✕ No	Standard datasets (ImageNet), reduces storage
PNG	Lossless (larger files)	✓ Yes	Segmentation masks, diagrams, high-fidelity data
BMP	Uncompressed	✕ No	Legacy systems, fast I/O but massive storage
TIFF	Lossless / Uncompressed	✓ Yes	Medical imaging, satellite imaging (high bit-depth)
WEBP	Lossy / Lossless	✓ Yes	Web scraping, modern efficient storage

Image Histograms

A histogram is a graphical representation of the **pixel intensity distribution** in a digital image.

- **X-axis:** Intensity levels (usually 0 to 255).
- **Y-axis:** Number of pixels at that specific intensity.
- **Uses:**
  - Analyzing brightness and contrast.
  - Thresholding & Segmentation (finding background vs object).
  - Histogram Equalization (enhancing contrast).



Why Preprocessing?

Raw data is messy, inconsistent, and mathematically unstable for Neural Networks.

Reduce Noise

Remove sensor artifacts and irrelevant background details.

Reduce Compute

Downsample to fit memory constraints (e.g., 224×224 for ResNet).

Improve Models

Normalize gradients for stable and faster backpropagation.

Raw Image → Preprocessing → CNN Model → Prediction

Common Preprocessing Operations

Standard operations using tools like OpenCV or PIL:

- **Spatial:** Resize, Crop, Padding.
- **Geometric:** Flip, Rotate, Affine transforms.
- **Filtering:** Blur (Gaussian), Sharpen, Edge detection.
- **Color:** Contrast adjustment, Grayscale conversion.

Example on left: Filtering operation enhancing edge frequencies (Sharpening).

Image Normalization

Neural networks struggle with large input values (like 255). Large inputs lead to large weights, causing vanishing/exploding gradients during training.

$$X_{norm} = \frac{X - \min(X)}{\max(X) - \min(X)} \approx \frac{Pixel_{Value}}{255.0}$$

1. Min-Max Scaling [0, 1]

Simple division by 255. Fast and standard for baseline models.

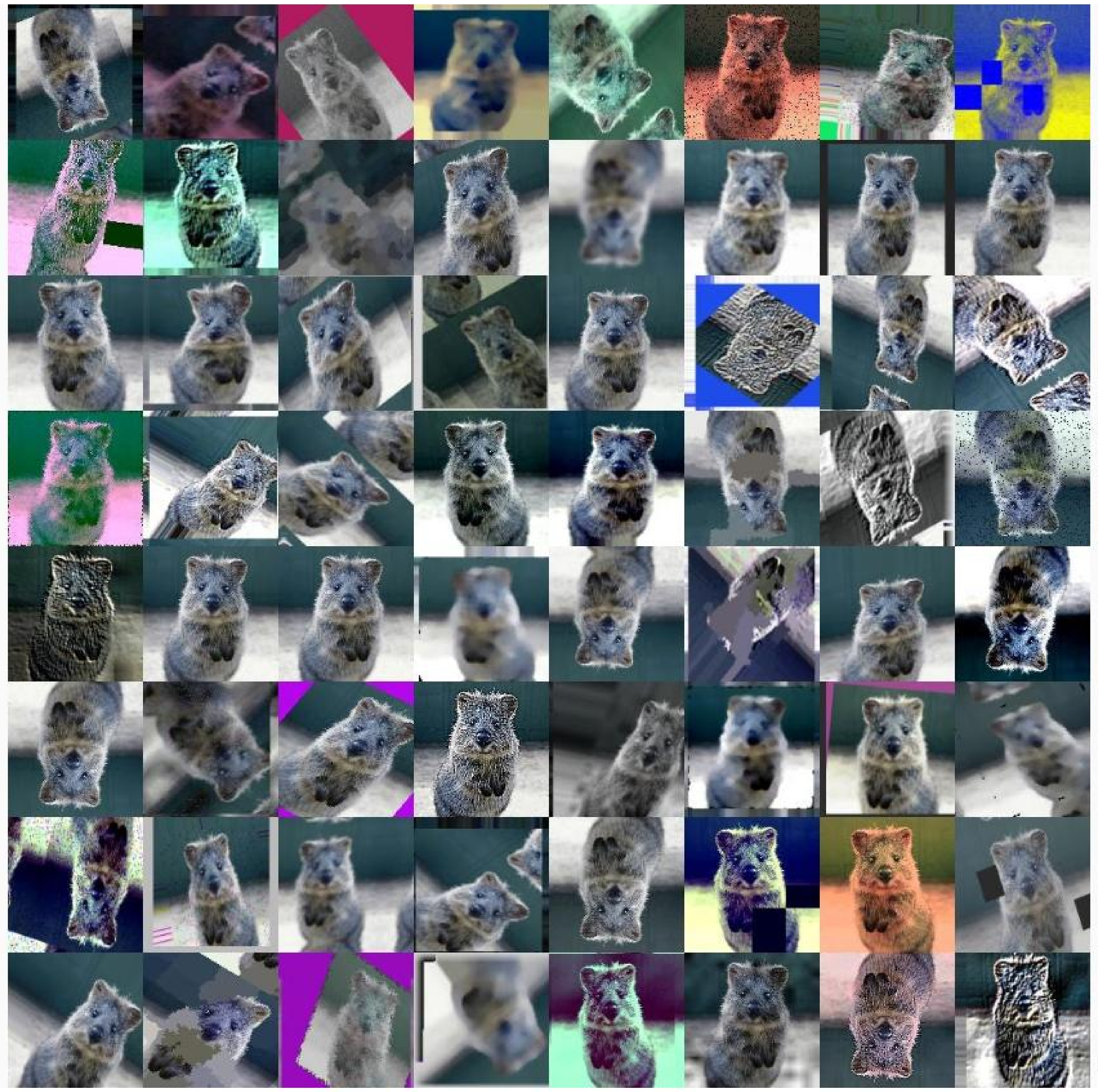
2. Standard Scaling (Z-Score)

Subtract dataset mean, divide by dataset standard deviation (e.g., ImageNet stats). Centers data at 0 with unit variance.

Data Augmentation

Artificially expanding the size of a training dataset by creating modified versions of images.

- **Why?** Prevents **Overfitting** and improves model Generalization.
- **Common Techniques:**
  - Random Rotations
  - Horizontal/Vertical Flips
  - Random Crop & Zoom
  - Color Jitter (Brightness/Contrast)
  - Gaussian Noise



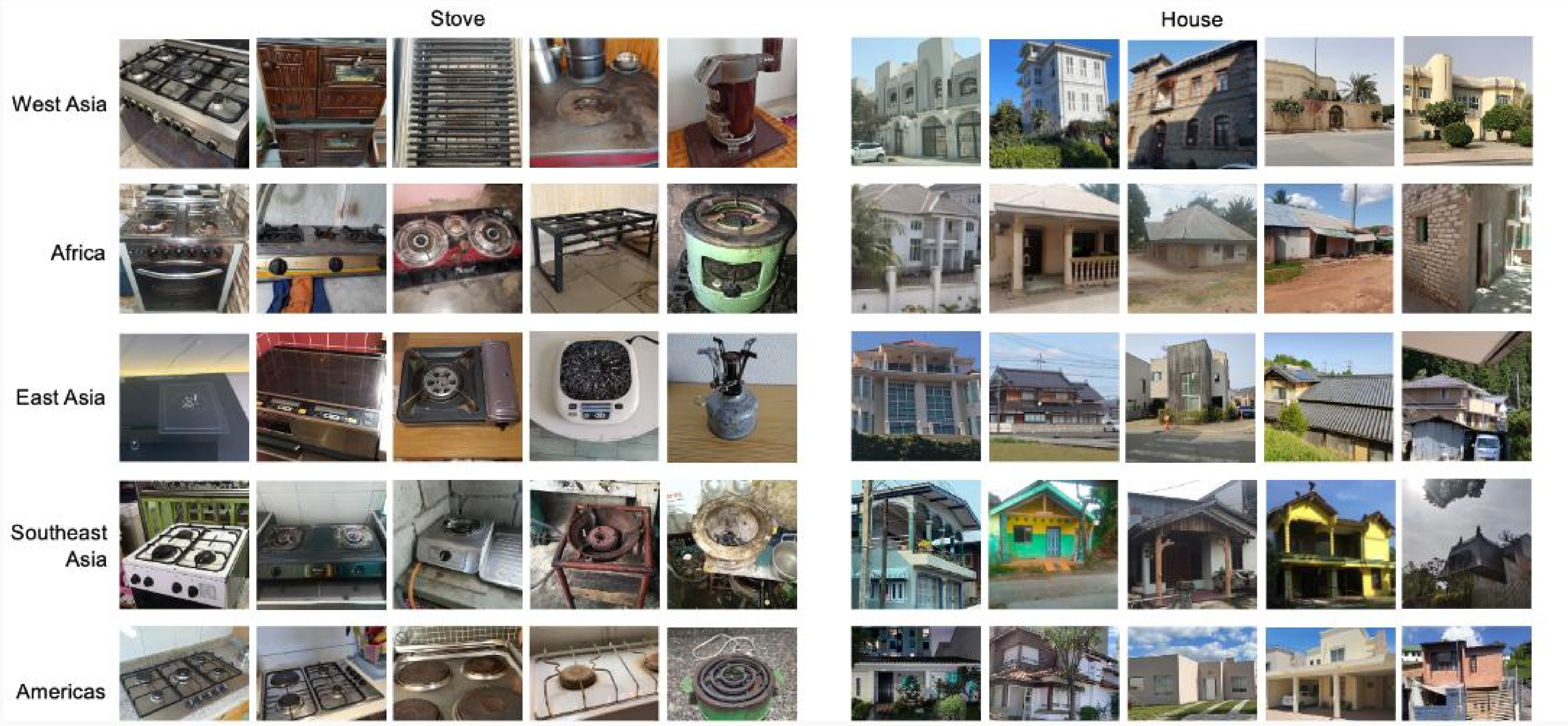
Real-World Pipeline (Training)

How a typical PyTorch/TensorFlow DataLoader prepares an image for a Convolutional Neural Network:

1. Load	Read JPG/PNG from disk into memory (H×W×3 matrix).
2. Resize / Crop	Force uniform dimensions (e.g., resize to 256×256, random crop to 224×224).
3. Augment	Apply random flips, rotations, or color jitter (Training only!).
4. To Tensor	Convert to Float32. Change shape from (H×W×C) to (C×H×W).
5. Normalize	Scale pixels to [0, 1] or apply Z-score normalization.

Interactive Activity: Think-Pair-Share

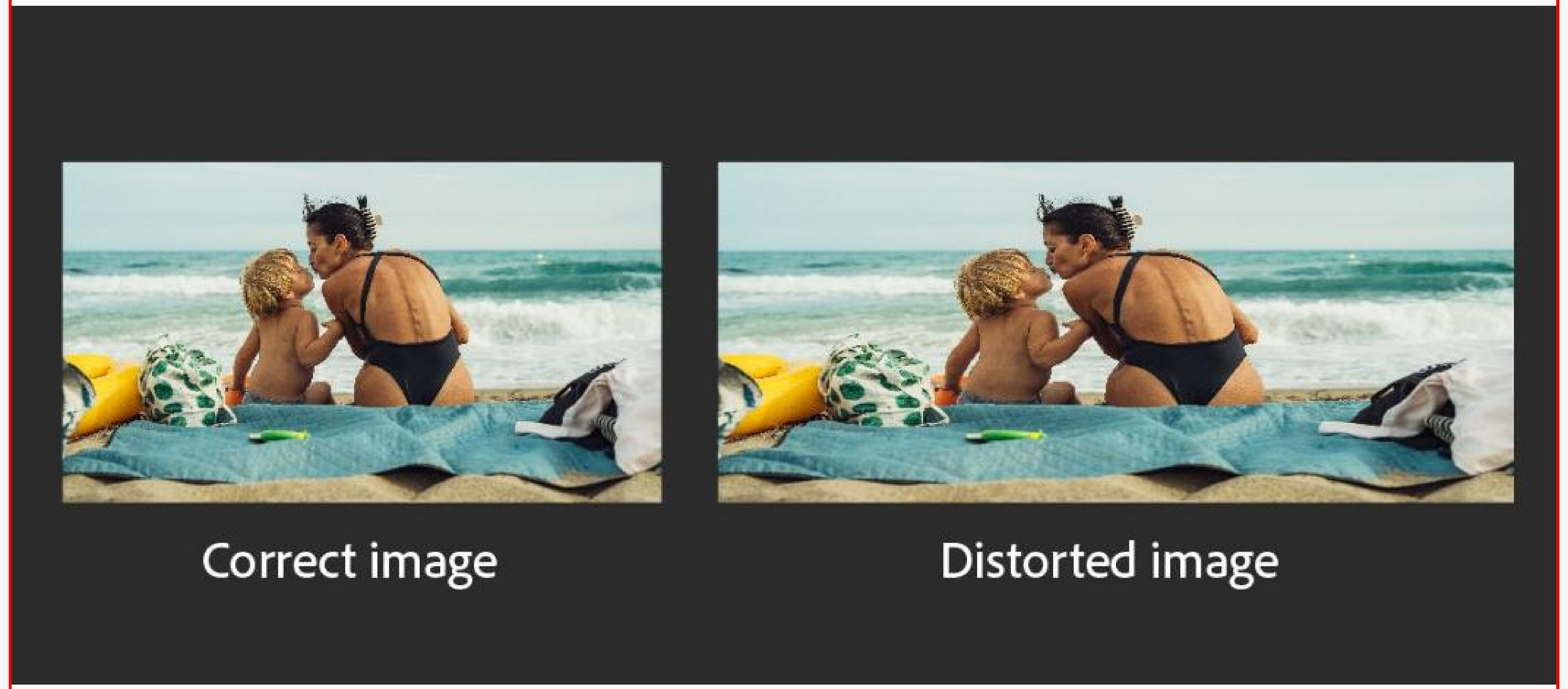
Discuss with your neighbor. For this diverse dataset sample, identify:



1. Likely Resolution variations?
2. Number of Channels?
3. Ideal Color Space for processing?
4. Top 3 Preprocessing steps required before feeding to a CNN?

Common Implementation Mistakes

- **RGB vs BGR:** Training in RGB but doing inference with OpenCV (BGR). Model will fail silently.
- **Forgetting Normalization:** Training loss explores to NaN, or model refuses to converge.
- **Aspect Ratio Distortion:** Blindly resizing (e.g., 1920×1080 to 224×224) squishes the image. (Use padding/cropping instead).
- **Data Leakage:** Applying data augmentation to your validation or test sets.



Example of destructive aspect ratio distortion

Lecture Summary

1. Images are Matrices

Digital images are discrete grids of pixels. Each pixel stores intensity values across 1 or more channels.

2. Representation Matters

Resolution affects spatial detail. Color spaces (RGB, HSV) change how information is mathematically grouped.

3. Preprocessing is Vital

Resizing, cropping, and normalization are mandatory steps to convert wild data into a mathematically stable tensor.

4. Augmentation Generalizes

Data augmentation creates artificial variance, preventing neural networks from memorizing the training set.

Homework & References

Homework Task

Download 5 diverse images from the internet.

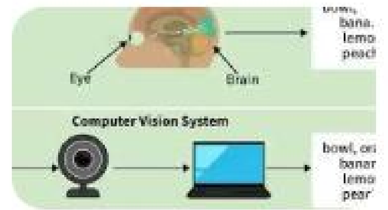
Write a Python script (using PIL or cv2) to output for each:

- Width, Height, and Channels
- Maximum and Minimum Pixel Values
- Display the normalized (0-1) tensor representation
- Plot the Grayscale Histogram

Reading & References

- **Textbook:** Richard Szeliski, *Computer Vision: Algorithms and Applications* (Chapter 2 & 3).
- **Documentation:** OpenCV Image Processing Tutorials.
- **Frameworks:** PyTorch `torchvision.transforms` docs.
- **Course Inspiration:** Stanford CS231n Lecture Notes.

Image Sources



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Source: www.geeksforgeeks.org

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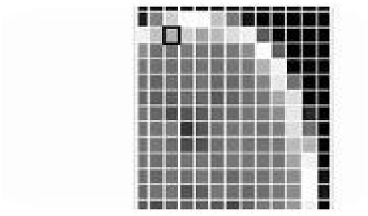
Source: www.istockphoto.com

https://www.brainline.org/sites/default/files/slides/mri.jpg

Source: www.brainline.org



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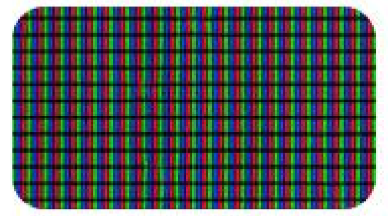
Source: cloverlearning.com



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Source: www.howtogeek.com

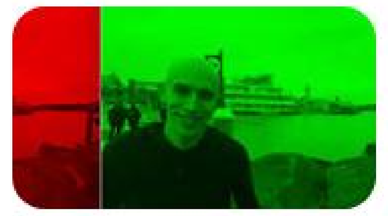
https://finerworks.com/App_Themes/finerworks/images/pixelated-blurry-okay.jpg

Source: support.finerworks.com

Image Sources

https://applescoop.org/image/wallpapers/mac/crystal-clear-blue-water-with-unique-rock-formations-beach-2025-best-ultra-hd-high-resolution-4k-desktop-backgrounds-wallpapers-for-mac-linux-and-windows-pc-11-03-2025-1741731303-hd-wallpaper.jpeg

Source: applescoop.org



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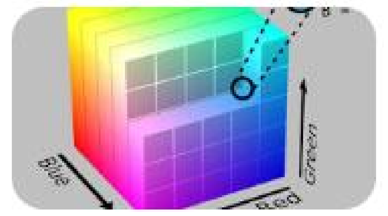
Source: pyimagesearch.com



https://upload.wikimedia.org/wikipedia/commons/8/83/RGB_Cube_Show_lowgamma_cutout_b.png

Source: alexanderell.is

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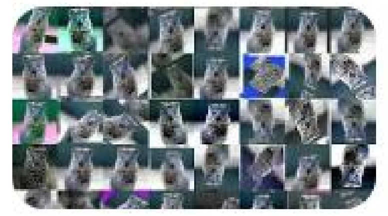


https://towardsdatascience.com/wp-content/uploads/2021/06/1Kd4PlhpU1l-tr1PK7WDsgQ.jpeg

Source: towardsdatascience.com

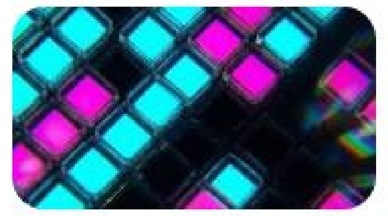
Image Sources

https://visualai.princeton.edu/geode/imgs/sample_images.png

Source: visualai.princeton.edu

https://helpx-prod.scene7.com/is/image/HelpxProd/pr-help-next-distortion?$pjpeg$&jpegSize=200&wid=1200

Source: helpx.adobe.com



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Source: stockcake.com